

LOGIC IS TWO

Journey Through the 5-Day Module: Design & Iteration Log

This document chronicles the iterative design process for our cooperative puzzle game, "**Logic is Two**". Over five distinct phases, we focused on refining player experience, discarding mechanics that didn't work, and ensuring the core asymmetrical co-op loop felt rewarding. Our primary goal was to evolve the game from a raw concept into a highly polished, player-centric experience.

Day 1: Establishing the Core Hook & Foundation

We set out to create a puzzle game that forced *true* cooperation. Instead of just two characters moving on a screen, we wanted their very abilities to be codependent.

- **What we added:** The asymmetrical block-pushing mechanic. Player 1 (Shadow) can only interact with Nouns, and Player 2 (Mist) can only interact with Properties. We also established the basic grid and elemental hazards (Fire, Water, Gaps).
- **Initial Scope:** We mapped out a massive 10-level campaign to test every possible interaction and edge case we could think of.
- **What we implemented for UX:** A time-rewind feature to undo mistakes instantly.

The Thinking Process: We knew that word-logic games can be brain-melting. By splitting the sentence structure between two players, they are forced to talk to each other to solve the puzzle. We added the Rewind feature on Day 1 because in a game about experimentation, punishing players by making them restart an entire level for one wrong push discourages creativity.

Day 2: Streamlining & Trimming the Fat

After evaluating our initial prototype, we realized our ambition was negatively impacting the pacing of the game.

- **What we removed:** We drastically cut the campaign down from 10 levels to 5 tightly designed stages. We removed repetitive puzzle designs that didn't teach the player anything new.
- **Terminology Change:** We replaced the property "FLOAT" with "JUMPER".
- **What we added:** A dedicated "How to Play" onboarding screen before the game starts.

The Iterative Leap: Quality over quantity. Ten levels felt exhausting, so we curated only the most interesting "Aha!" moments into five distinct challenges. We also noticed "FLOAT" was slightly ambiguous for crossing gaps, whereas "JUMPER" intuitively told players exactly what the rule allowed them to do. Finally, we realized players were jumping into the game without understanding the asymmetric controls, so the onboarding screen became a priority.

Day 3: UI/UX Refinement & Visual Hierarchy

The game was mechanically sound, but reading the board state was confusing. Players needed to know their status at a glance.

- **What we added:** A structured UI layout with a distinct side panel. This panel clearly lists the characters' unique traits, controls, and dynamically displays the "Active Synergy" (the rules currently formed on the board).
- **What we refined:** Feedback systems. We added dynamic text that lights up when a player steps on a trigger button.

The Thinking Process: Puzzle games are about processing information. If the player has to constantly scan the board to remember if "MIST IS SAFE" is active, cognitive load is wasted on the UI rather than the puzzle. Centralizing the active rules in a sidebar solved this. The trigger feedback was added because players need immediate confirmation that their actions (like standing on a button) had a meaningful impact on the environment.

Day 4: Atmosphere and "The Juice"

A logic game can easily feel dry or academic. Day 4 was entirely about injecting "juice"—making the game feel alive, reactive, and rewarding to interact with.

- **What we added:** Environmental animations (flickering fire, flowing water, bobbing gaps, and idle breathing animations for the characters).
- **Visual Rewards:** A moving starfield background to give depth, a VHS-style glitch effect when players use the Rewind feature, and a confetti explosion upon level completion.

The Iterative Leap: Mechanics make a game playable, but aesthetics make it memorable. We thought about how solving a hard logic puzzle feels—it's a moment of relief and celebration. Adding the confetti directly rewards that feeling. The VHS rewind effect was added to give the time-travel mechanic a tactile, nostalgic feel, making a "mistake" feel like a cool feature rather than a failure.

Day 5: Accessibility & Final Polish

For the final iteration, we watched how players physically interacted with the game on a single keyboard.

- **What we added:** A "Swap Controls" toggle button on the main menu, allowing players to instantly switch who uses WASD and who uses the Arrow keys.
- **Competitive Metrics:** We added a Move Counter and a Time Counter.
- **What we removed/optimized:** We stripped out complex, experimental audio synthesis code in favor of highly stable, traditional audio triggers to prevent browser compatibility issues.

The Thinking Process: In local co-op, physical comfort is key. Sometimes the person sitting on the right wants to play Shadow, but Shadow is hardcoded to WASD (the left side of the keyboard). Adding the control toggle was a direct response to making the game physically comfortable to play. We added the timers and move counters as an afterthought for replayability—once a player solves the puzzle, the counters give them a reason to play it again, optimizing their logic for a speedrun.